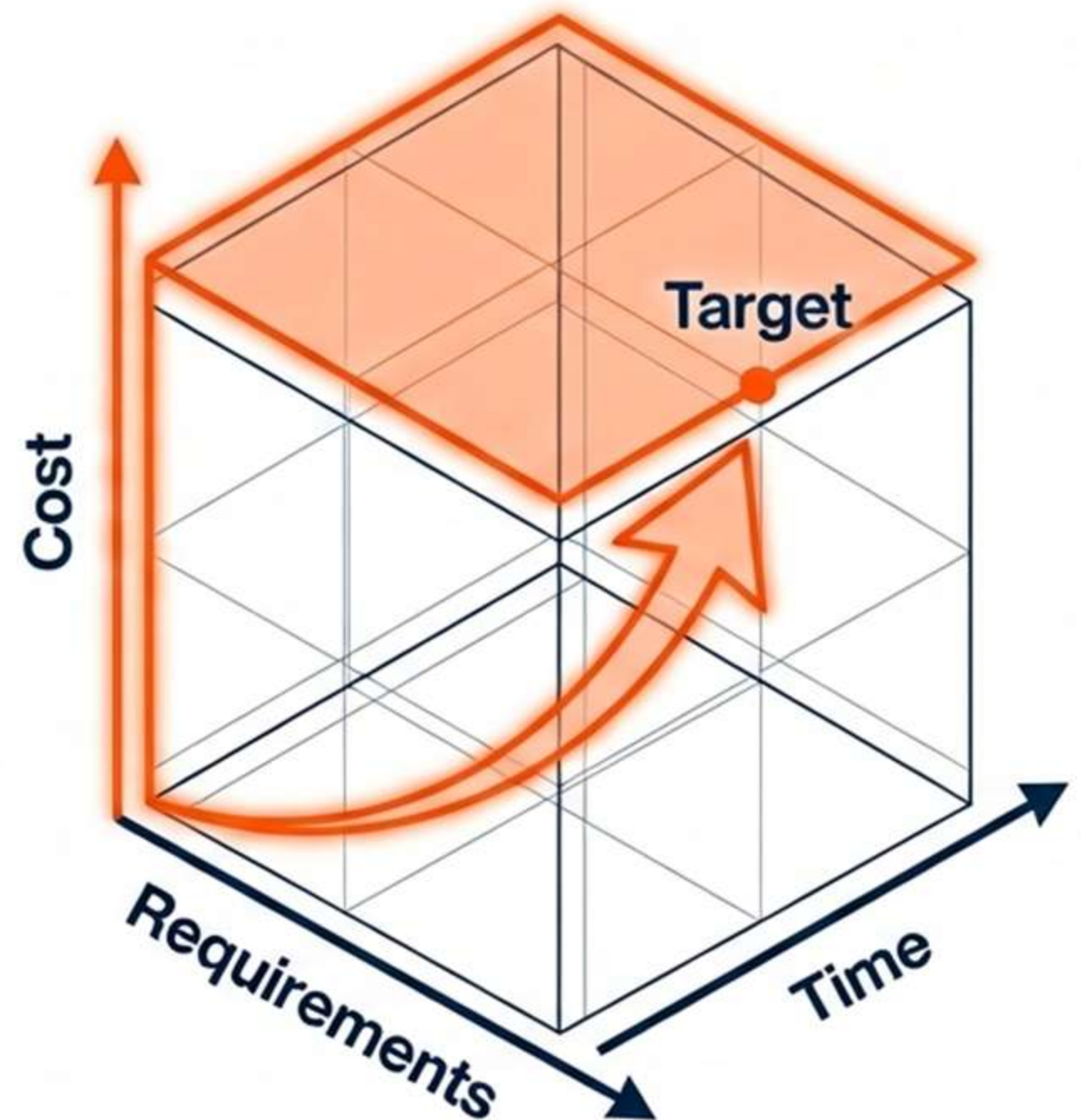




Cost Estimating & Budgeting

INS3044 IT Project Management – Topic 5
Lecturer: Dr. NGUYEN The Loc



WHY THIS MATTERS

CLO 2: IMPLEMENT management of cost.

CLO 8: IMPLEMENT overall project plans.

CLO 9: DEVELOP integrity and responsibility (Ethics).



The Guesstimate



	A	B	C	D
1				
2	Item	Cost	Percentage	Timeline
3		\$10,500	25%	Q3 2024
4		\$10,000	40%	Q3 2024
5		\$10,000	25%	Q3 2024
6				
7				
8				
9				
10				
11				

The Baseline

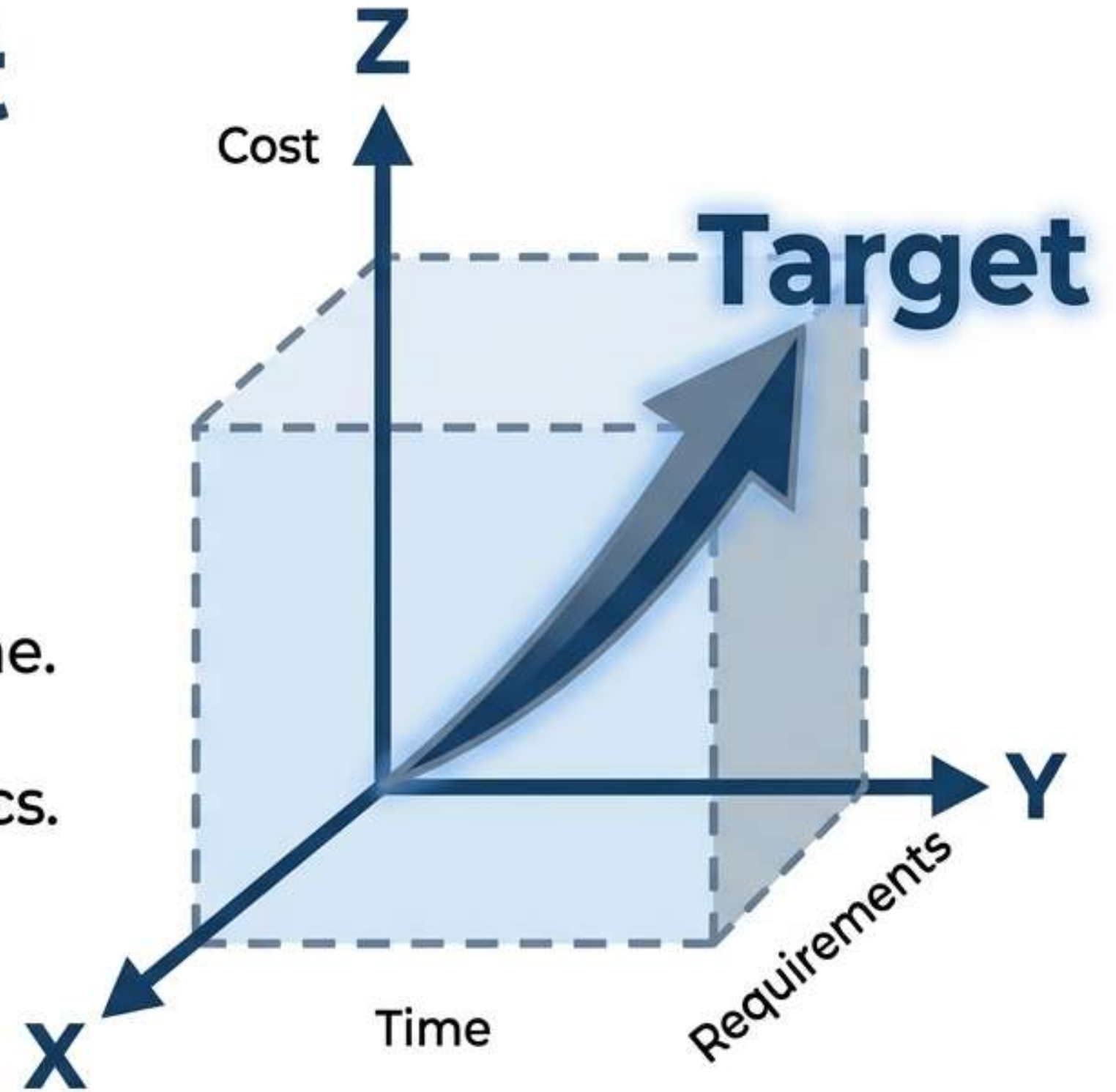
Learning Objectives

- ✓ Define estimating and budgeting.
- ✓ Analyze cost escalation factors.
- ✓ Apply four estimating methods.
- ✓ Structure a time-phased budget.
- ✓ Understand project cost accounting.



The Floating Target

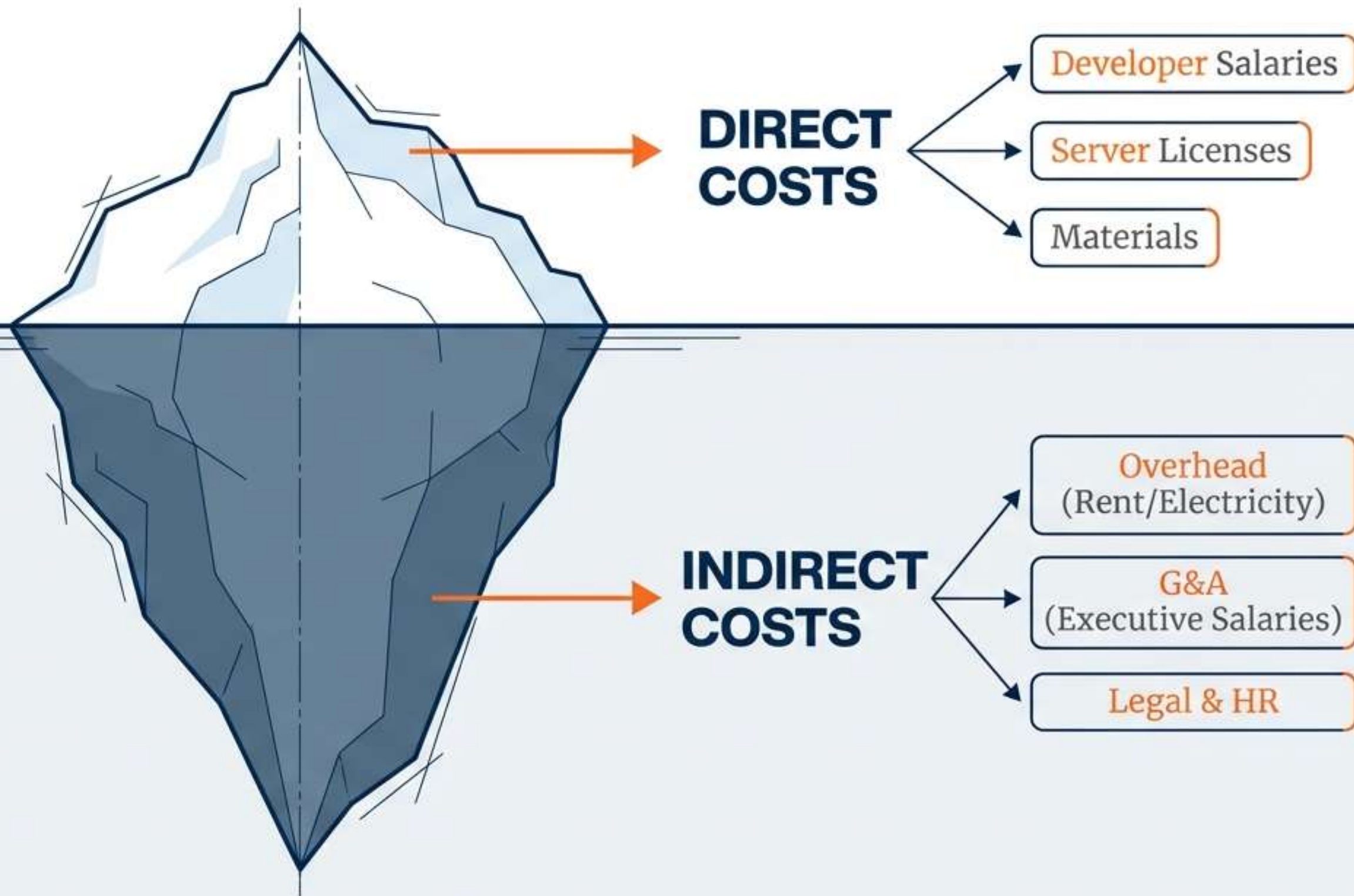
- ✓ Projects balance three main goals.
- ✓ Hit the targeted budget cost.
- ✓ Finish within scheduled project time.
- ✓ Meet all required performance specs.
- ✓ Goal: Hit this 3D target.



Merrweather
Service Now Display
(#4A4A4A)

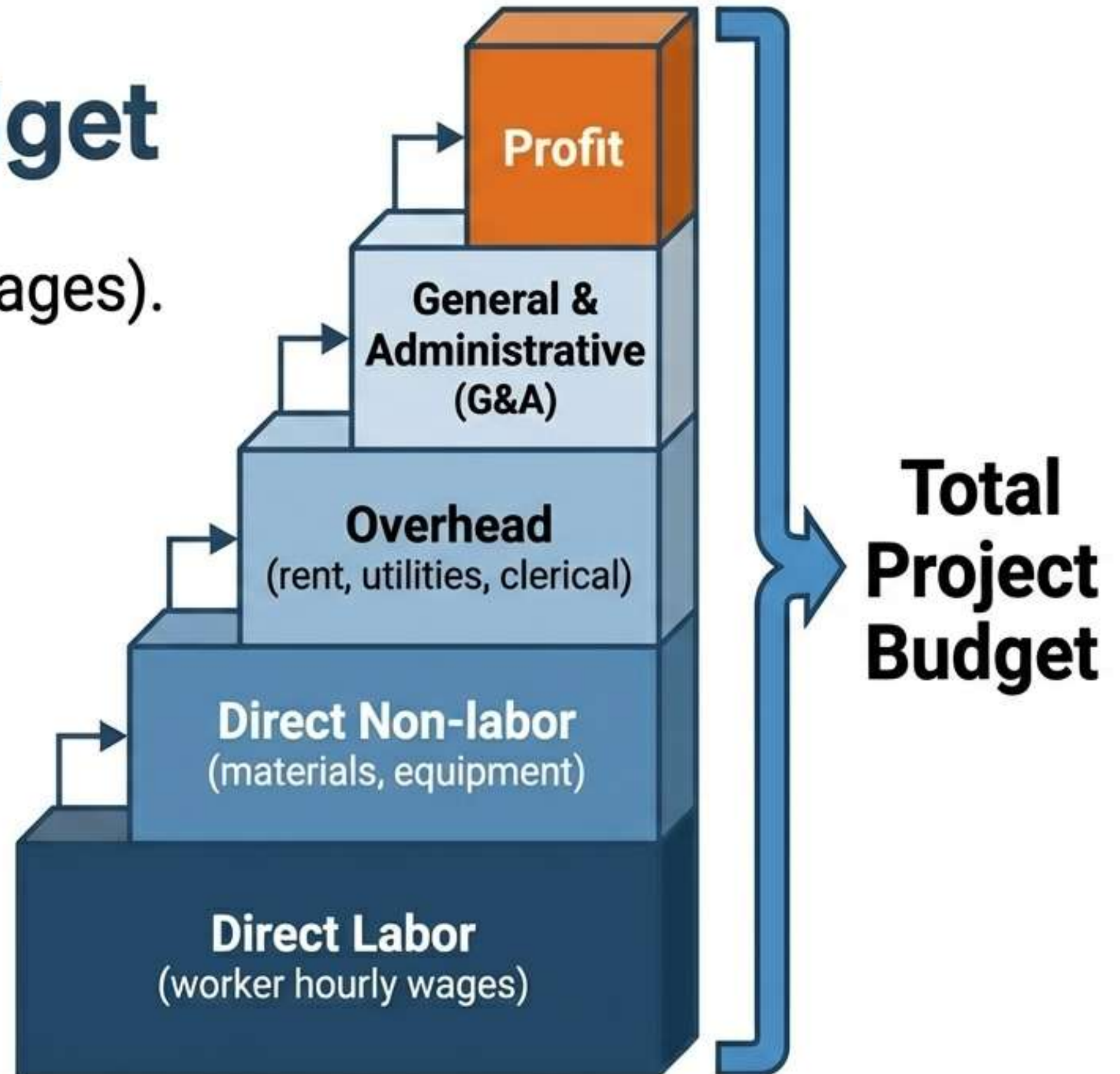
COST = Resources Expended
PRICE = Cost + Profit

THE VOCABULARY OF MONEY



Elements of a Budget

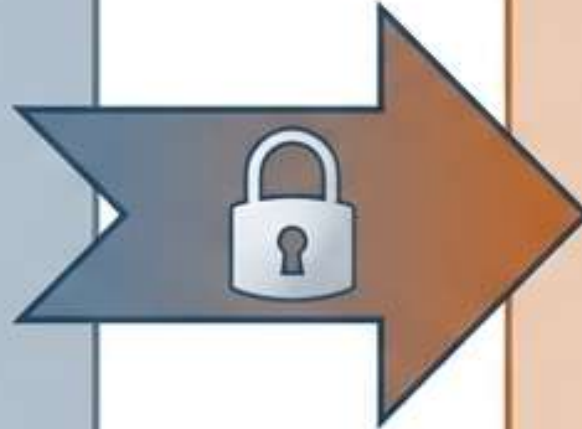
- Direct labor (worker hourly wages).
- Direct non-labor (materials, equipment).
- Overhead (rent, utilities, clerical).
- General and administrative (G&A).
- Profit and total billing.
- Combined into final project budget.



Estimates vs. Budgets

Estimate:

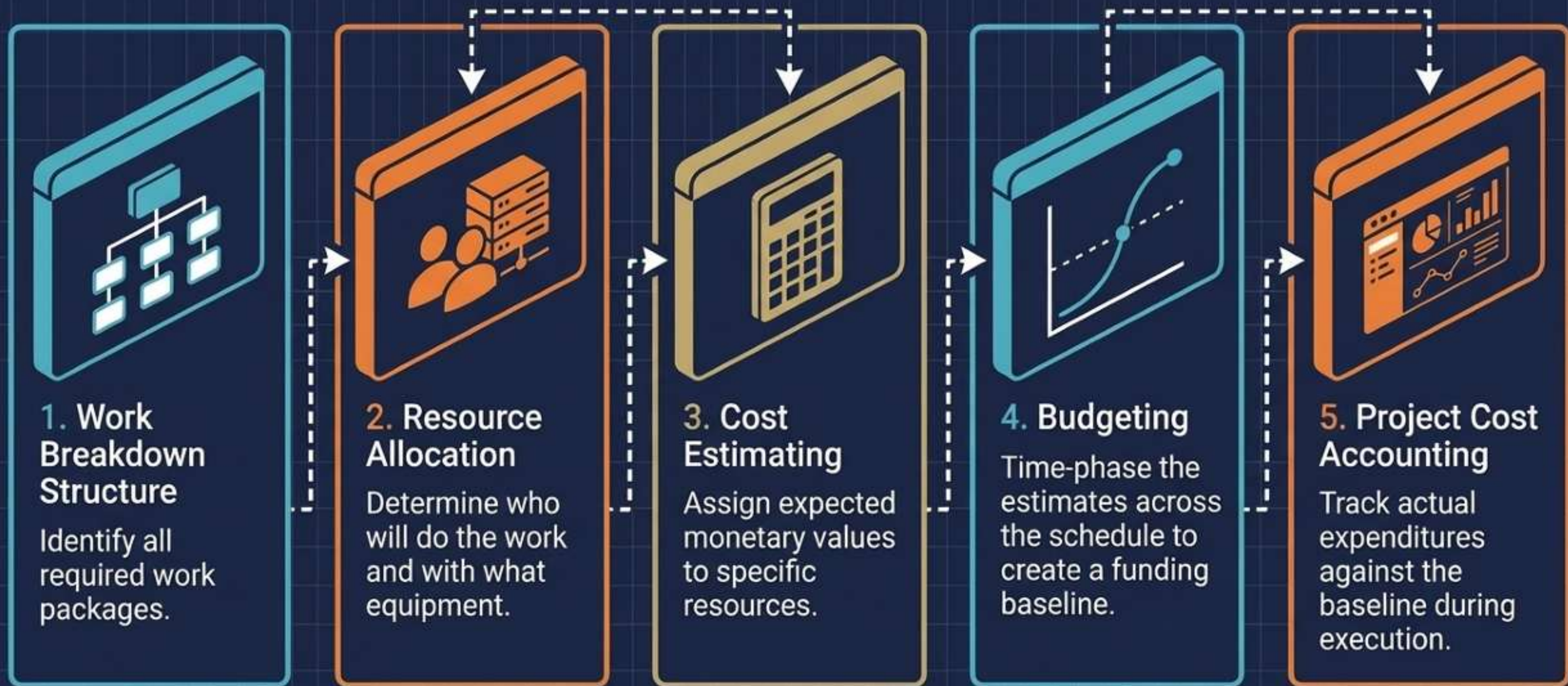
- Made before work begins.
- A prediction of future costs.
- Often revised as details emerge.



Budget:

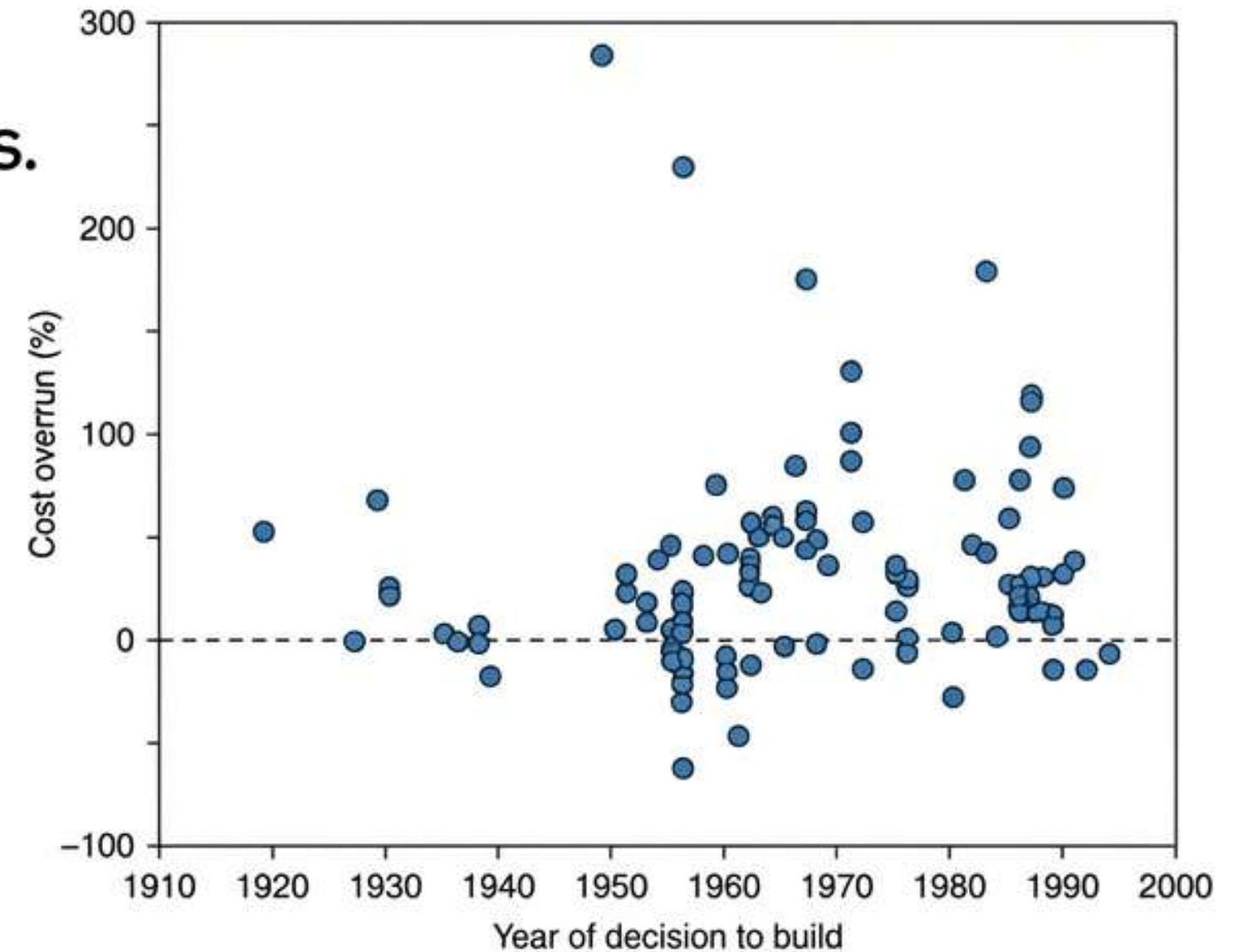
- The approved baseline cost.
- A firm financial commitment.
- Used to measure actual performance.

From Scope to Budget

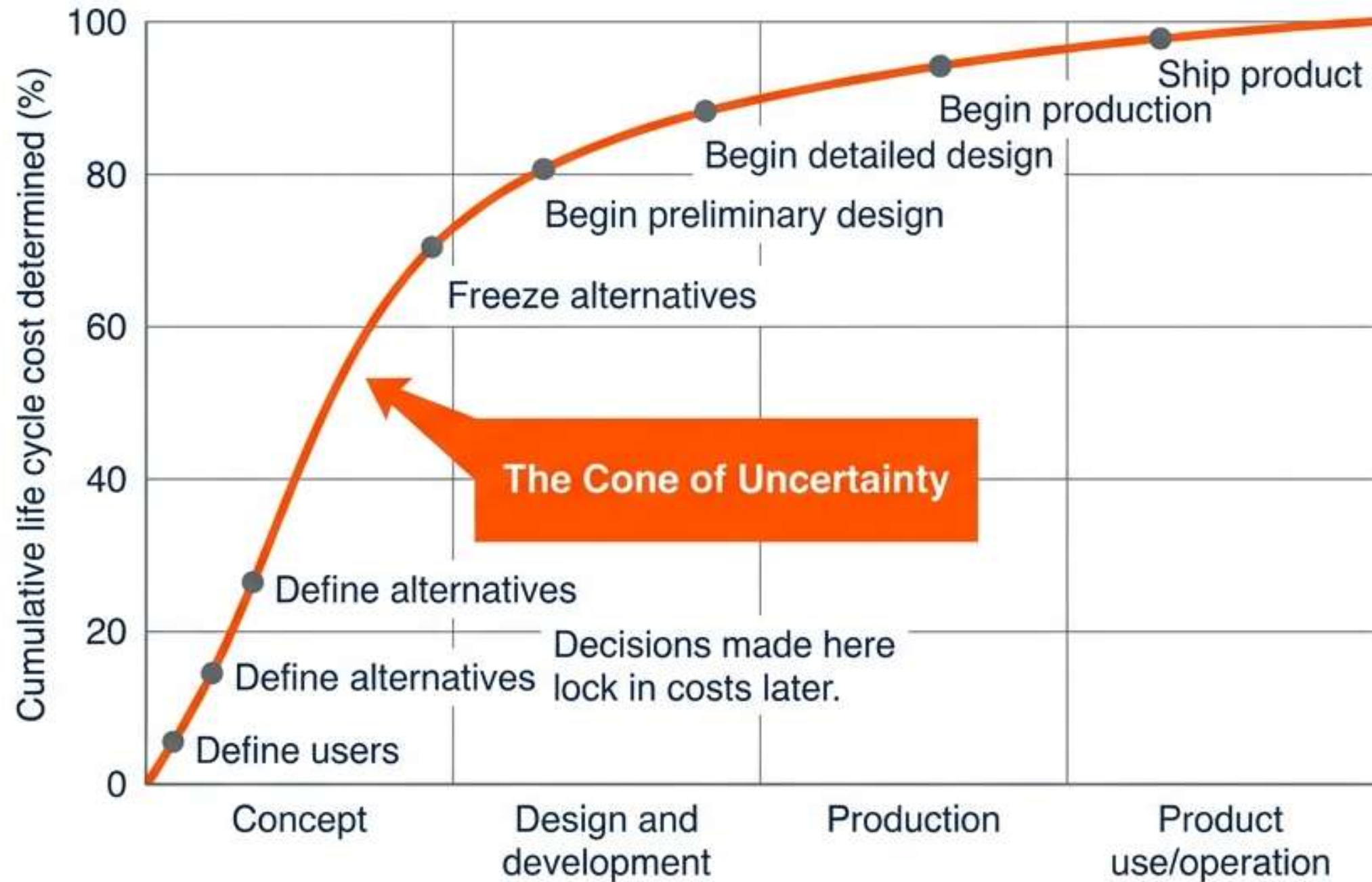


The Cost Escalation Problem

- ✓ Actual costs often exceed estimates.
- ✓ Early projects lack detailed information.
- ✓ Estimators show extreme over-optimism.
- ✓ "Buy in" deliberately lowers bids.
- ✓ Escalation is a normal risk.



THE PROBLEM: COST ESCALATION



Uncertainty

Scope Creep

Inflation

Unethical "Buy-in"

The Cone of Uncertainty Narrows Over Time

You cannot have a perfect estimate on day one. Estimate accuracy is a function of the project life cycle.

Conception Phase

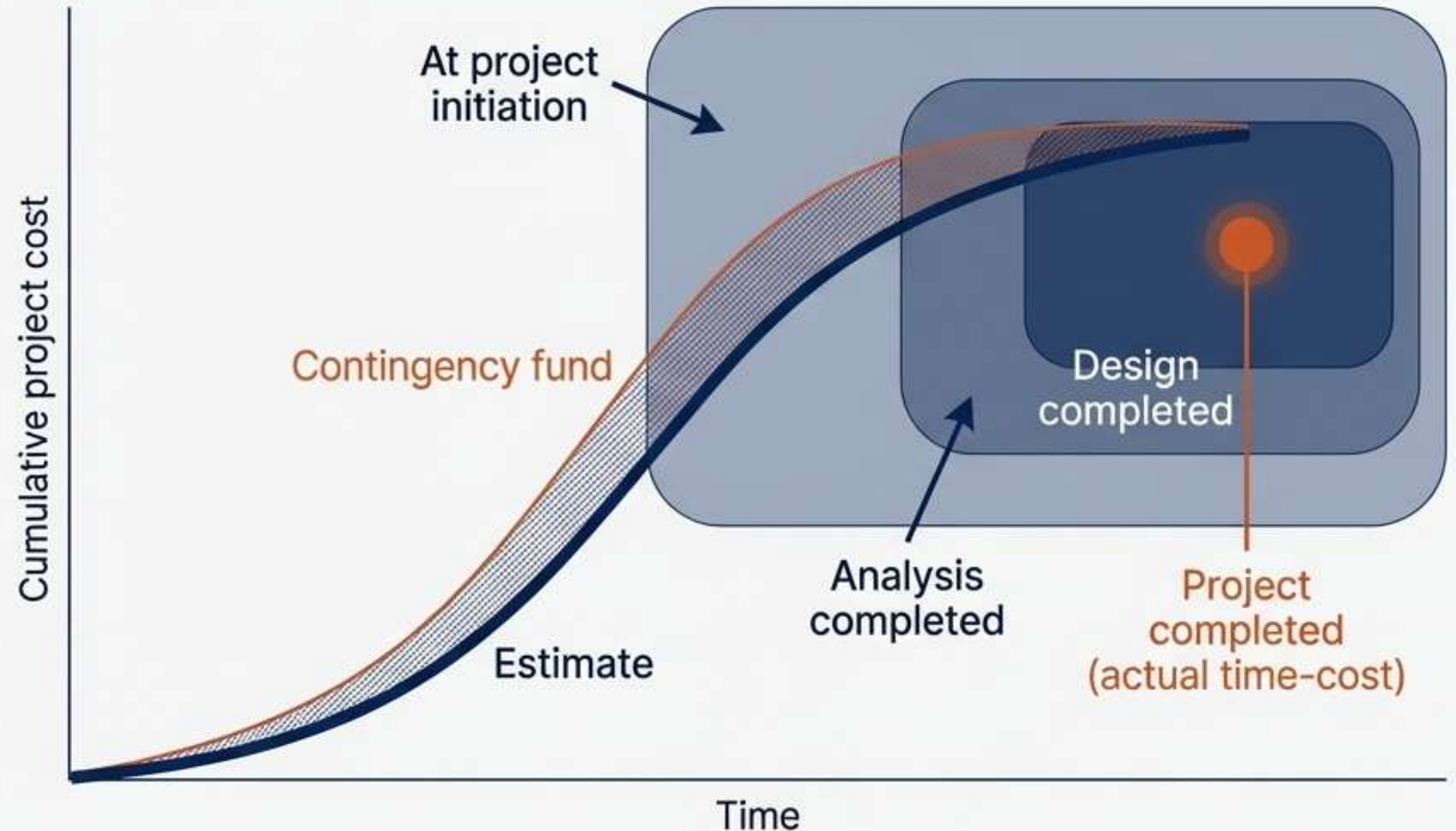
The 'region of uncertainty' is massive. Estimates are largely guesstimates based on high-level standards.

Definition Phase

As the system and work activities are nailed down, material costs and labor times become clearer.

Execution Phase

The baseline is locked. Changing decisions here is incredibly costly due to sunk costs.



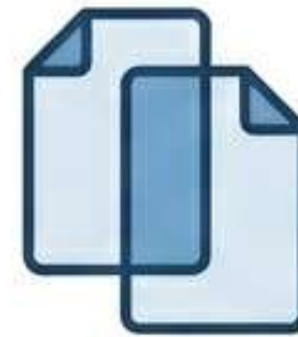
Four Estimating Methods

Top-Down



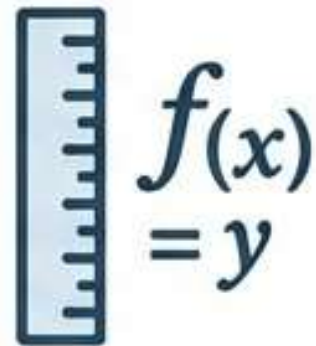
Expert Judgment

Uses expert experience.



Analogy

Compares to past projects.



Parametric

Uses math and scaling.



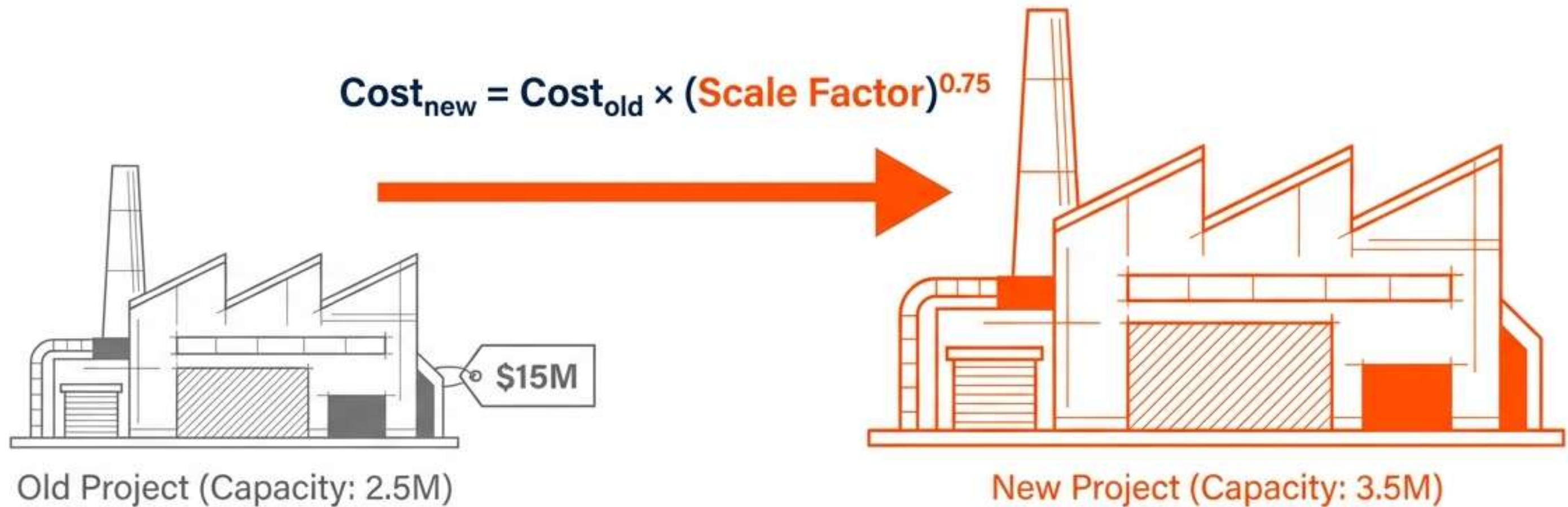
Cost Engineering

Detailed task analysis.

Bottom-Up

METHOD 1: TOP-DOWN APPROACHES

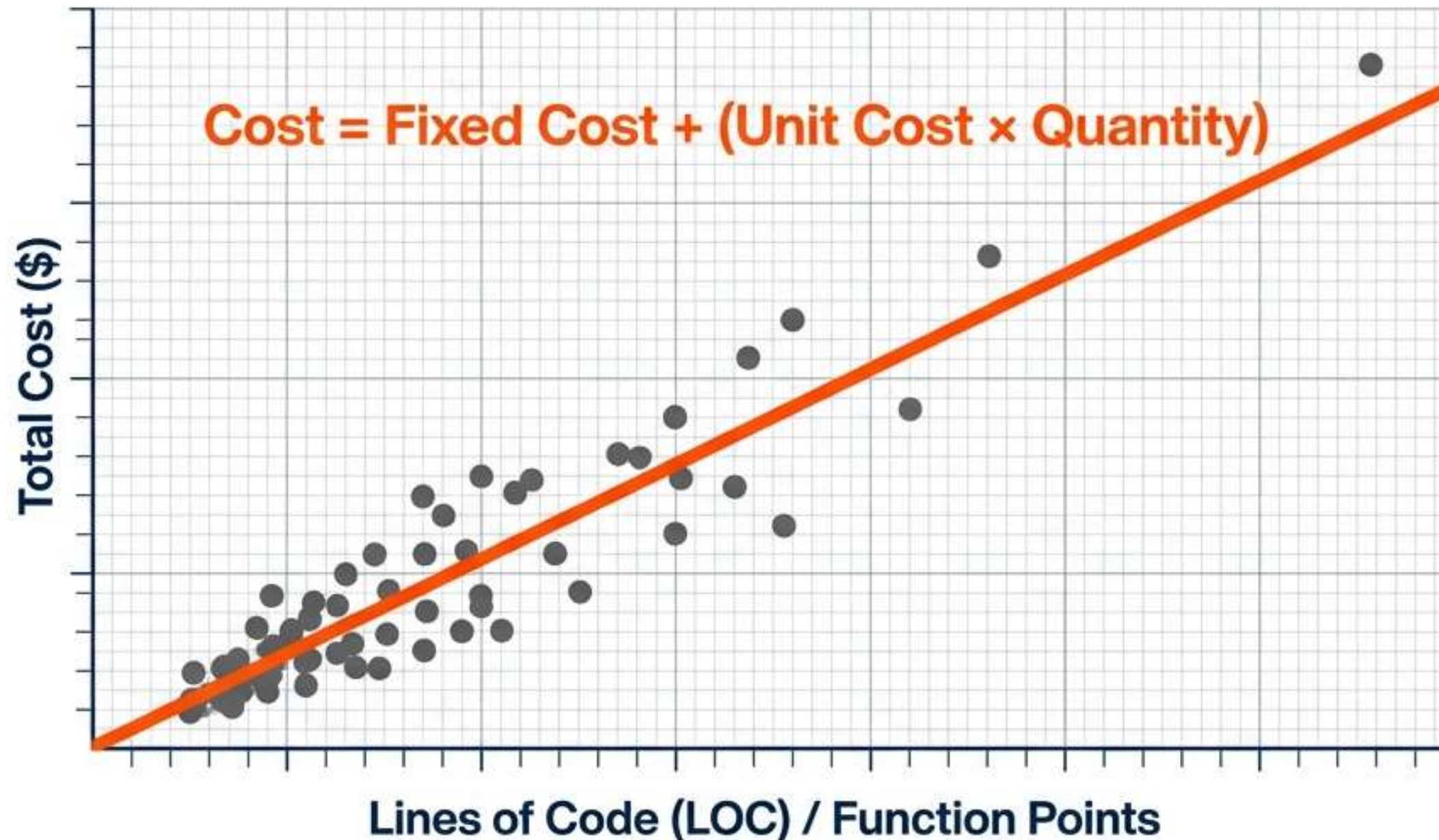
Analogy & Scaling



Estimated Cost $\approx$ \$18.8M

METHOD 2: PARAMETRIC ESTIMATING

The Math of Prediction



EXAMPLE CALCULATION

Parameter: 100

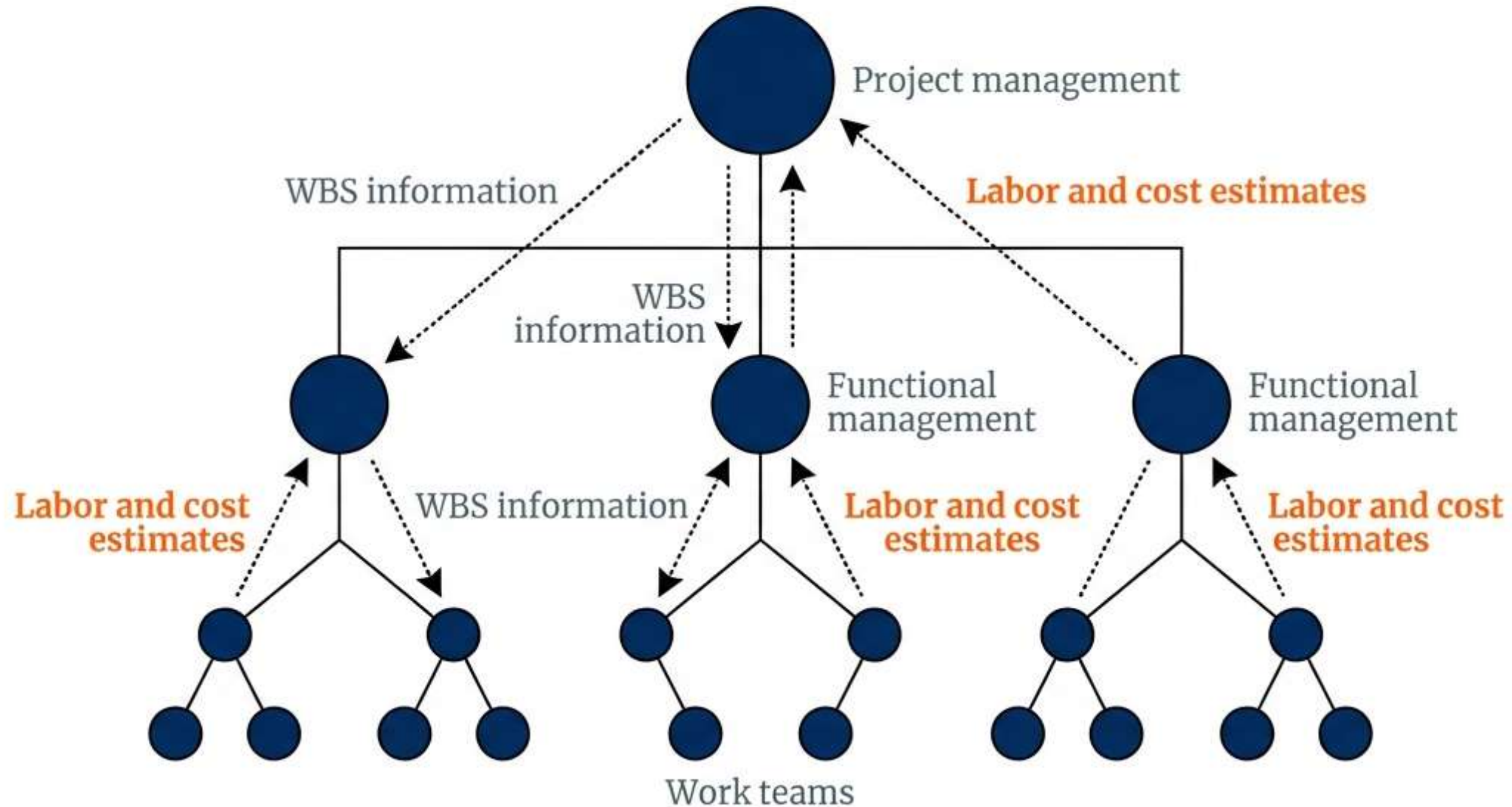
Function Points

Rate: \$500/Point

Total = \$50,000

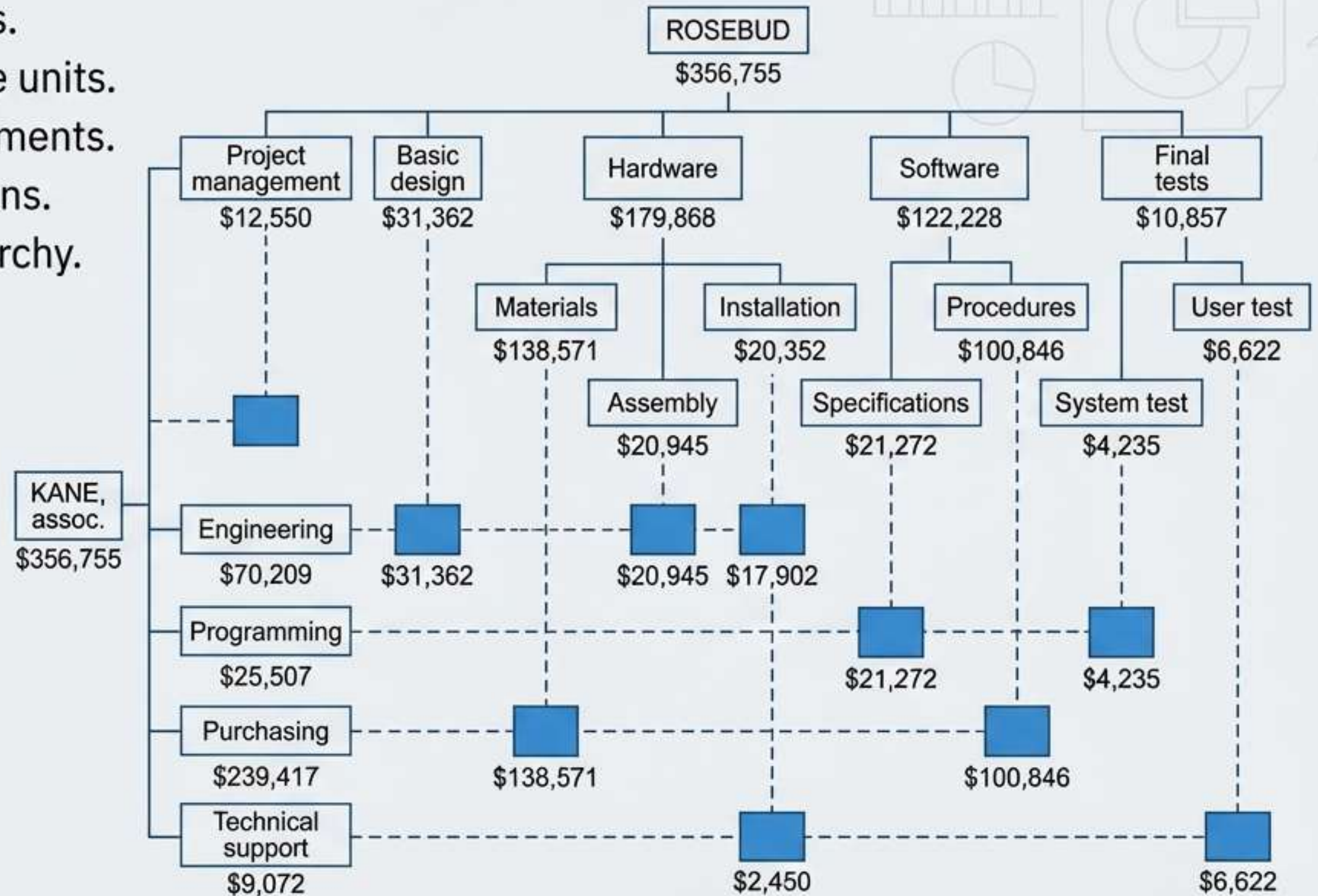
METHOD 3: BOTTOM-UP (COST ENGINEERING)

Detailed Data from the Ground Up



Control Accounts & WBS

- Budgets map to WBS packages.
- Divide project into manageable units.
- Assign costs to specific departments.
- Allows exact tracking of overruns.
- Forms clear cost account hierarchy.



Time-Phased Budgets

- Budgets must include time schedules.
- Spreads costs over project months.
- Shows distribution of budgeted costs.
- Predicts exact cash flow needs.
- Compares monthly budgets versus actuals.

Work package: (W + X) Final Tests									
Charge	Rate	Months						Totals	
		1	2	3	4	5	6	Hours	Cost
Direct labor									
Professional	\$35/hour					30	30	30	1,050
Associate	\$30/hour					90	90	90	2,700
Assistant	\$20/hour								0
Direct labor cost							3,750		3,750
Labor overhead	75%						2,813		2,813
Other direct cost*						1,200	2,107		3,307
Total direct cost						1,200	8,670		9,870
General/administrative	10%					120	867		987
Total costs						1,320	9,537		10,857

*Should be itemized to include costs for materials, freight, subcontracts, travel, and all other nonlabor direct costs.

Accuracy vs. Precision

- **Accuracy** means closeness to actuals.
- Precision means exact decimal points.
- Goal: Maximize actual estimate accuracy.
- Use PERT three-point estimates.
- Combines optimistic, pessimistic, and likely.



Precise but
Inaccurate

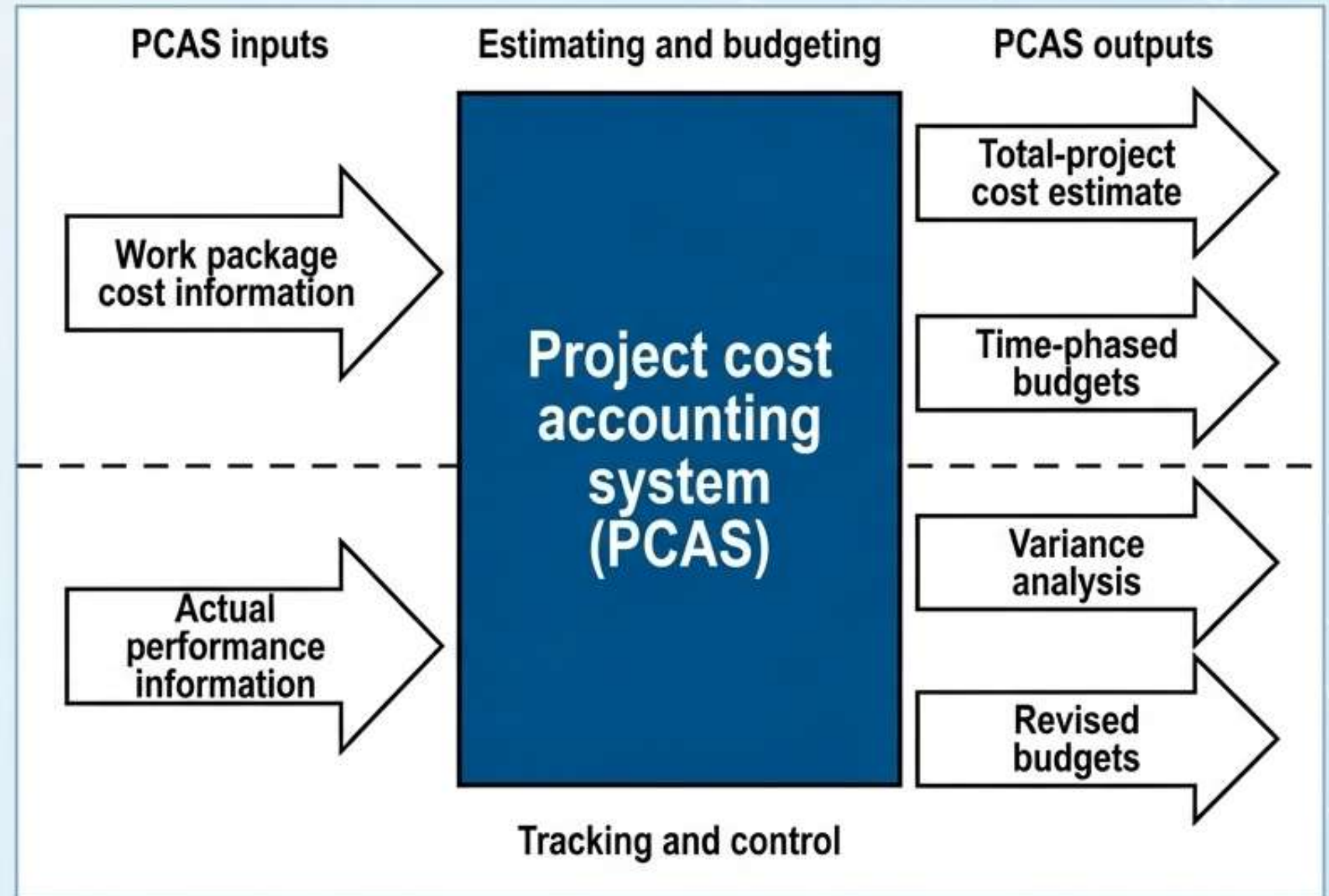
$$CE = \frac{a + 4m + b}{6}$$



Accurate and
Precise

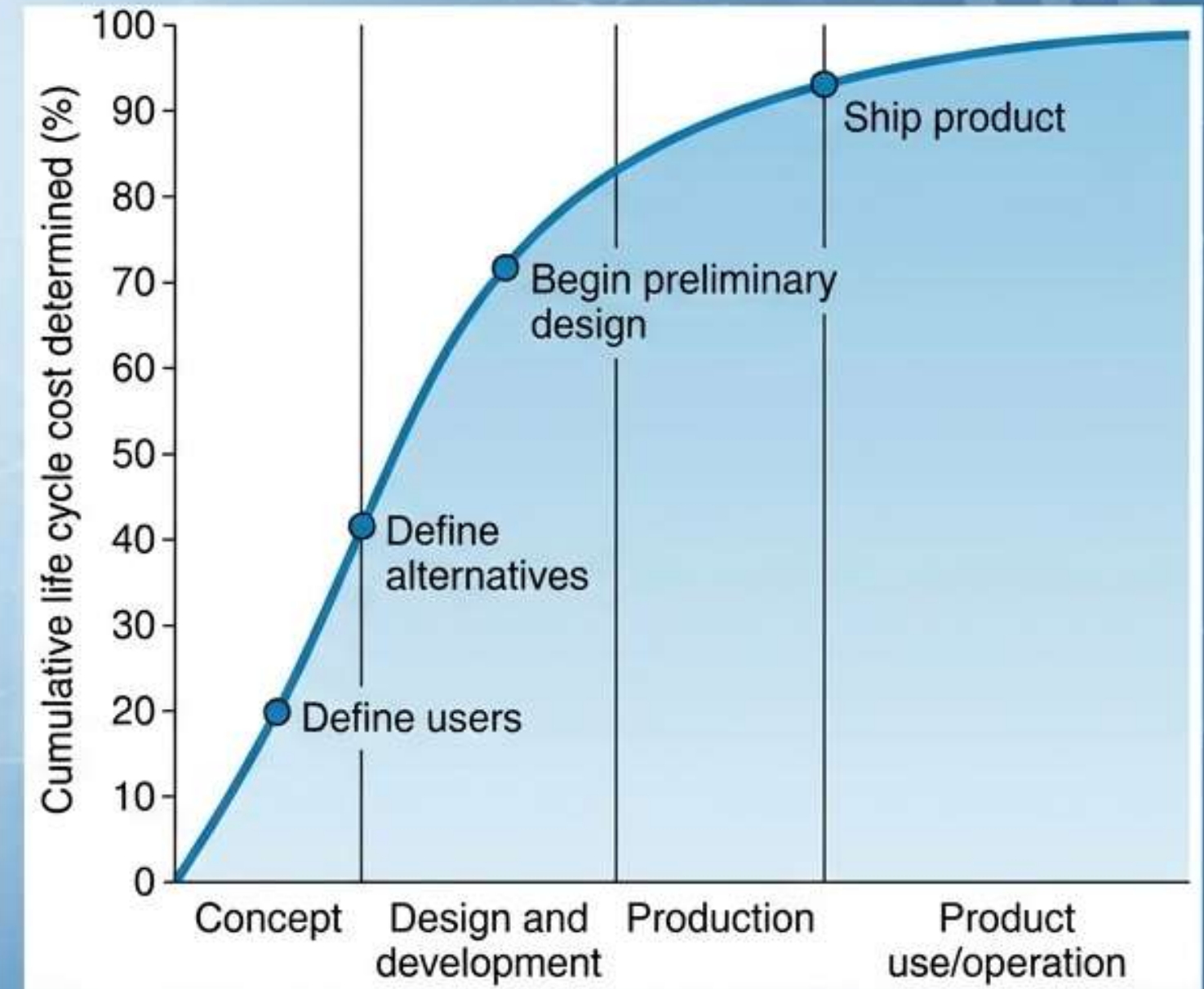
Project Cost Accounting Systems

- PCAS tracks all project costs.
- Aggregates work package cost estimates.
- Tracks actual versus budgeted costs.
- Quickly identifies any cost variances.
- Essential for large IT projects.



Life Cycle Costs (LCC)

- Looks beyond project execution phase.
- Cradle-to-grave system costs.
- Includes operations and maintenance costs.
- Maintenance costs often exceed project.
- Decisions today impact future costs.



Takeaways & Class Practice



- Estimates predict; budgets control spending.
- Choose appropriate cost estimating methods.
- PCAS tracks budgets against actuals.



- Which estimating method is best?
- How do you prevent escalation?
- Why track life cycle costs?

Exercise: The Cafe App Estimate

10 Minutes

Work Package (Task)	Hours	Rate (\$50/hr)	Total
Subtotal			
+ 20% Overhead			
GRAND TOTAL			

1. Pick 3 Tasks.
2. Estimate Hours.
3. Calculate Cost.
4. Add Overhead.